

Statistical Methods for Data Science (Fall 2025)

Mini Project 4

Instructions:

- Due date: Dec1, 2025.
- Total points = 20.
- Submit a typed report.
- Justify all steps and provide all relevant explanations.
- You can work on the project either individually or in a group of no more than two students. In the case of the latter, submit only one report for the group and include a description of the contribution of each member.
- It is OK to discuss the project with other students in the class (even those who are not in your group), but each group must write its own code and answers. If the submitted report (including code and answer) is similar (either partially or fully) to someone else's, this will be considered evidence of academic dishonesty, and you will be referred to appropriate university authorities.
- You must use the following template for your report:

Mini Project #

Name

Names of group members (if applicable)

Contribution of each group member

Section 1. Answers to the specific questions asked

Section 2: R code. Your code must be annotated. No points may be given if a brief look at the code does not tell us what it is doing.

1. (12 points) Consider the data stored in the file **VOLTAGE.CSV** on eLearning. These data come from a Harris Corporation/University of Florida study to determine whether a manufacturing process performed at a remote location can be established locally. Test devices (pilots) were set up at both the remote and the local locations and voltage readings on 30 separate production runs at each location were obtained. In the dataset, the remote and local locations are indicated as 0 and 1, respectively.
 - (a) (3 points) Perform an exploratory analysis of the data by examining the distributions of the voltage readings at the two locations. Comment on what you see. Do the two distributions seem similar? Justify your answer.
 - (b) (6 points) The manufacturing process can be established locally if there is no difference in the population means of voltage readings at the two locations. Does it appear that the manufacturing process can be established locally? Answer this question by constructing an appropriate confidence interval. Clearly state the assumptions, if any, you may be making and be sure to verify the assumptions.
 - (c) (3 point) How does your conclusion in (b) compare with what you expected from the exploratory analysis in (a)?

2. (8 points) The file **VAPOR.CSV** on eLearning provide data on theoretical (calculated) and experimental values of the vapor pressure for dibenzothiophene, a heterocycloaromatic compound similar to those found in coal tar, at given values of temperature. If the theoretical model for vapor pressure is a good model of reality, the true mean difference between the experimental and calculated values of vapor pressure will be zero. Perform an appropriate analysis of these data to see whether or not this is the case. Be sure to justify all the steps in the analysis.